

Matgeo-2.10.8

AI25BTECH11019-Menavath Sai Sanjana

Question

Let $(b) = 4\hat{i} + 3\hat{j}$ and (c) be two vectors perpendicular to each other in the XY plane. All vectors in the same plane having projections 1 and 2 along (b) and (c) , respectively, are given by .

Solution

Consider $(r) = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$, where r_1 and r_2 are the x- and y-components of the vector we want to find.

$$(b) = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \quad (c) = \begin{pmatrix} x \\ y \end{pmatrix} \text{ (perpendicular to } (b) \text{)} \quad (1)$$

$$\text{Scalar projection on } (b) : \frac{(r)^T (b)}{\|(b)\|} = 1 \quad \Rightarrow \quad (r)^T (b) = 5 \quad (2)$$

$$\text{Since } (c) \perp (b), \quad 4x + 3y = 0 \quad \Rightarrow \quad x = -3t, y = 4t \quad (3)$$

$$\text{Scalar projection on } (c) : \frac{(r)^T (c)}{\|(c)\|} = 2 \quad \Rightarrow \quad (r)^T (c) = 10t \quad (4)$$

Solution (Continuation)

$$(r)^T (b) = 4r_1 + 3r_2 = 5 \quad (5)$$

$$(r)^T (c) = -3r_1 + 4r_2 = 10 \quad (6)$$

$$\text{Solving the system: } 4r_1 + 3r_2 = 5, \quad -3r_1 + 4r_2 = 10 \quad (7)$$

$$r_1 = -\frac{2}{5}, \quad r_2 = \frac{11}{5} \quad (8)$$

Conclusion

Conclusion: The required vector is $(r) = \begin{pmatrix} -\frac{2}{5} \\ \frac{11}{5} \end{pmatrix}$.

Graphical Representation

